Resource-Constrained Language Modeling with QK-Normalized Rotary Attention and Born-Again Distillation

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Abstract

This report studies a small autoregressive language model under a strict CPU inference budget. The submitted model is a six-layer, six-head, width-348 decoder with per-head query/key RMS normalization, rotary position embeddings, a SwiGLU feed-forward network, and a strictly causal within-window cache. Two first-generation students learn from a pair of existing teachers; a second-generation student learns from the two first-generation students. Each new student starts from random weights and trains for 19,200 updates. The final exported model obtains 1.450151 validation bits per UTF-8 byte (BPB) and 1.468030 full-test BPB, compared with 2.071084 and 2.101260 for the course baseline. Its measured validation CPU time is 4.701 times the fastest adjacent baseline run, peak resident memory is 1.652 GiB, and complete inference assets occupy 34,743,281 bytes. A matched direct-distillation control has slightly better validation BPB (1.448343) but misses the conservative five-times CPU threshold in its recorded bracket. The paired experiment does not support a quality gain from the extra distillation generation. We report the resulting trade-off and the limits of causal attribution explicitly.

1. Task, protocol, and selection rule

The task is next-token prediction on the supplied WikiText-2 text with the fixed BPE-2048 tokenizer and independent causal windows of 256 tokens. The course evaluator and data files are unchanged. We report bits per byte as BPB = total negative log-likelihood in nats / (ln(2) x UTF-8 byte count). The validation split contains 376,599 scored targets and 1,148,007 UTF-8 bytes; the full test split contains 428,405 targets and 1,292,013 bytes. A lower BPB is better. All formal scores use the supplied evaluator with CPU FP32 and four PyTorch threads.

Model development, checkpoint choice, cache calibration, and resource screening used training data and validation data only. The selected method and checkpoint fingerprint were frozen before the initial full-test score. A later independent test rerun checked reproducibility and did not change the model or settings. The course limits are at most five times baseline CPU scoring time, 4 GiB peak evaluation RAM, and 64 MiB of complete inference assets. The submission includes a directly loadable checkpoint; retraining is unnecessary for scoring.

2. Method

2.1 Decoder architecture

The final student has six decoder blocks, width 348, six attention heads of width 58, feed-forward width 688, and 7,934,052 learned parameters. It uses tied input/output token embeddings, bias-free projections, pre-attention and pre-feed-forward RMSNorm, causal scaled-dot-product attention, SwiGLU feed-forward layers, and residual dropout 0.05. The architecture builds on Transformer attention [1], rotary position embeddings [2], RMSNorm [3], and SwiGLU [4].

Each block applies a separate RMSNorm to the query and key vectors within each head before rotary rotation and attention. We call this QK-Norm for brevity, but it is an RMSNorm-based variant: the original Query-Key Normalization paper [5] uses L2 normalization and a learned attention scale. Therefore the report does not claim an exact reproduction of that paper's layer. The six-head width-384 option was screened earlier and exceeded the course's five-times CPU budget; width 348 was used for the completed training runs.

2.2 Two-generation distillation

The two retained starting teachers produce next-token probability distributions. First-generation students S29 and S31 each learn from the arithmetic mean of those teachers' probabilities. The second-generation student S37 learns from the arithmetic mean of S29 and S31. A matched direct student D37 learns from the original teachers. S37 and D37 use the same architecture, seed 37, input sampling schedule, 19,200-step budget, and calibration grid; teacher lineage is the intended experimental difference. This extends the born-again idea of successive student-teacher generations [6], while adding a two-teacher probability mixture.

For student distribution p and frozen teacher mixture q, the per-token training objective is L = 0.5 CE(y, p) + 0.5 CE(q, p), with teacher temperature 1. The member probabilities are averaged after softmax, rather than averaging logits. Each student starts from a fresh random initialization. Training uses AdamW, batch size 32, context 256, FP32, weight decay 0.05, gradient clipping at 1, a peak learning rate of 0.001, 100 warmup steps, and cosine decay to 0.0001. Four late checkpoints at steps 17,760, 18,240, 18,720, and 19,200 are accumulated in FP64 and averaged for an FP32 export.

The experimental pipeline is:

Figure 1. The born-again route adds two first-generation students before S37; D37 shares the same final student recipe and learns directly from the original pair.

2.3 Causal cache and CPU export

At inference, the model combines temperature-scaled neural probabilities with a within-window cache inspired by continuous neural caches [7]. The cache compares normalized hidden states and assigns probability mass to tokens already observed at strictly earlier positions in the same window. It has no cross-window state, no access to future tokens, and no stored validation or test answers. The first prediction in each window uses only the neural distribution. The final validation-chosen settings are neural temperature 1.1, cache similarity scale 13, and cache mixture weight 0.05.

The CPU export uses complex multiplication for pairs of rotary dimensions instead of separate real-valued temporary tensors. It does not change learned weights or the architecture. The full validation BPB is identical before and after this implementation change (1.4501509920851057). Cache settings were selected on validation from 28 predeclared points: three neural temperatures, three similarity scales, three nonzero mixture weights, and one zero-cache point.

3. Experimental design and cost

3.1 Baseline and paired control

The fixed course baseline is a four-layer, four-head, width-128 GPT with 1,088,256 parameters. It trains from random weights for 1,200 steps and reaches 2.071083639 validation BPB and 2.101260438 test BPB.

The main paired comparison is S37 versus D37. Each sees 19,200 x 32 x 256 = 157,286,400 training targets, uses seed 37 and the same student model, and is evaluated under the same validation protocol. The two first-generation students add 314,572,800 training targets to the born-again route. The direct control therefore tests whether the extra teacher generation is worth its added training cost. It does not isolate QK-Norm, width, cache, or checkpoint averaging, which are shared by the paired students.

3.2 Training and search cost

All four new students together process 629,145,600 training targets. On the recorded NVIDIA GeForce RTX 3060 Laptop GPU, their update times total 24,826.9 seconds (6.90 hours), with another 766.3 seconds (0.21 hours) spent in in-loop validation. The final S29-S31-S37 route accounts for 5.19 update-hours; the direct D37 control accounts for 1.71. These measurements exclude the two starting teachers, which were trained in earlier experiments, and exclude other archived exploratory runs. The reported cost is therefore a lower bound on total project search and training.

The checkpoint search compares late individual checkpoints and their average on validation. Each candidate then uses the same 28-point cache grid. The CPU resource screen brackets candidate scoring with baseline runs on the same machine; the ratio uses the faster of the two adjacent baseline times. This conservative denominator avoids accepting a candidate only because one baseline run happened to be slow.

4. Results

Table 1. Course baseline and selected final model under the fixed full-split scorer. CPU time and RAM are from the final candidate's adjacent validation bracket on an Intel i7-12700H; the baseline time is the faster of the two adjacent runs.

Metric	Course baseline	Selected final model
Depth / width / heads	4 / 128 / 4	6 / 348 / 6
Validation BPB (lower is better)	2.071084	1.450151
Full-test BPB (lower is better)	2.101260	1.468030
Validation scoring time	6.096 s	28.657 s
CPU time ratio	1.000 x	4.701 x
Validation peak resident RAM	1.567 GiB	1.652 GiB

The final model reduces validation BPB by 0.620933 (29.98%) and full-test BPB by 0.633230 (30.14%) relative to the course baseline. Its inference bundle is 34,743,281 bytes, below the 67,108,864-byte limit. The recorded CPU ratio and peak resident RAM are also below their respective limits. These before-after gains combine changes in architecture, model width, training duration, distillation, checkpoint averaging, and cache calibration; they are not an isolated estimate of any one component.

Table 2. Equal-budget teacher-lineage comparison. "Default-cache" uses the same in-training cache settings for all students; "calibrated CPU" uses each candidate's validation-selected settings and the optimized inference implementation.

Student	Teacher source	Neural validation BPB	Default-cache BPB	Calibrated CPU BPB
S29	Original pair	1.477217	1.449056	-
S31	Original pair	1.482423	1.452582	-
S37	S29 + S31	1.488795	1.457722	1.450151
D37	Original pair	1.480347	1.453208	1.448343

The direct control beats the second-generation student by 0.004514 BPB with identical default-cache settings and by 0.001808 BPB after separate validation calibration. Thus the measured ablation does not show a predictive-quality benefit from the extra generation. S37 was selected because its conservative CPU ratio was 4.701, while D37's recorded ratio was 5.366. The final method's 1.450151 validation BPB also improves over the historical previous best of 1.468614 by 0.018463.

5. Critical analysis and reproducibility

The selection has a narrow performance margin and should not be interpreted as evidence that born-again training makes inference faster. S37 and D37 have the same model structure and inference code. D37's baseline-after run took 8.282 seconds, much slower than its 6.134-second baseline-before run; using the mean of those two baselines would give a 4.566 ratio instead of 5.366. This variation makes the pass/fail decision sensitive to the conservative timing rule and system load. More repeated CPU brackets would improve confidence, but the frozen submission uses the recorded fastest-baseline rule.

This study also lacks a paired no-QK-Norm or width-348 control at the full 19,200-step budget. The baseline comparison therefore cannot attribute the large BPB gain specifically to QK-Norm. The cache calibration is selected on one validation split and may not generalize equally to new corpora. Two first-generation students and prior starting teachers make the born-again route more expensive than direct distillation; the slight quality loss in the matched ablation is a meaningful negative result.

Reproducibility evidence includes the frozen selection record, source and checkpoint SHA-256 values, CPU resource logs, the original full-test result, and a later independent full-test rerun. Both test runs yield exactly 1.4680303500941403 BPB, with identical loss values for all 1,674 windows. The final checkpoint SHA-256 is 981d85a72df78ffa0a53845f1e00f2ddf0334c40682afaa7ad0d476274def681. Installation, four-stage training, validation scoring, and direct checkpoint evaluation commands are in README.md; the relevant machine-readable records are under reports/born-again-qknorm348-19200/. The original test was scored only after validation selection was frozen, and the later rerun was solely a reproducibility check.

Table 3. Main artifacts retained for independent checking. Report files are under `reports/born-again-qknorm348-19200/` *unless another location is shown.*

Artifact	What it verifies
`frozen-selection.json`	Candidate and cache settings fixed before full-test scoring
`optimized-summary.json`	Validation BPB and adjacent CPU resource bracket
`selected-test.json`	Original full-test score for the frozen candidate
`reproduced-test.json`	Independent repeat of the full-test score
`born-again-fast/checkpoint.pt`	Final weights identified by the reported SHA-256
`README.md`	Installation, training, and evaluation commands

6. Conclusion

A compact QK-normalized rotary decoder with two-generation distillation meets the recorded CPU, RAM, and asset limits while substantially improving on the course baseline's BPB. The controlled experiment shows that the extra distillation generation is not responsible for a quality gain relative to direct distillation at equal student budget. The report's practical outcome is a reproducible, loadable final model and an explicit account of the quality-cost trade-off, rather than a claim of a new standalone architectural effect.

References

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